

Bui Minh Hieu

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PROFESSIONAL SUMMARY

Junior Software Engineer with 2 years of experience in Fullstack Web development and AI-integrated applications. Proficient in building responsive user interfaces (UI), developing API backends and ML core including data collection, model training, optimization, deployment. Hands-on experience building production-grade ML features for face verification and mask detection. Familiar with 3D point cloud processing, ONNX deployment, and integrating ML models into production systems. Able to work across ML, backend, and frontend components in production environments.

WORK EXPERIENCE

AI Automation - MegaX

Oct 2025 – Now

- **ZaloOA AI-Chatbot:** Built AI-driven automation workflows for enterprise Zalo Official Account (Zalo OA) using n8n as the main orchestration platform. Integrated Zalo OA Webhook with n8n to process incoming messages, trigger scenarios, manage user sessions, and handle message routing without custom backend servers. Designed multi-step chatbot flows (FAQ, product recommendation, booking, customer onboarding, lead qualification) using n8n nodes, conditional logic, and data transformations. Connected n8n with Google Sheets, Google Drive, and Google Calendar APIs to store leads, retrieve product data, manage customer information, and automate reporting. Integrated LLM APIs (OpenAI/ChatGPT) with n8n to enhance natural language understanding and generate dynamic responses for different business use cases. Collaborated with enterprise clients to optimize response accuracy, reduce manual workload, and automate customer service operations on Zalo OA.

AI Engineer - Emage Development Co

Mar 2023 – Dec 2024

- **Automatic check-in system:** Built full ML pipelines including dataset collection, labeling workflow, augmentation, training, evaluation, and model deployment. Implemented and maintained a Face Verification system for attendance tracking, achieving 95% verification accuracy and 40% faster inference after model optimization. Trained and fine-tuned multiple architectures: ResNet, EfficientNet, MobileNetV2, InceptionV3. Applied OpenCV for preprocessing, normalization, detection, and video stream handling. Developed an internal web dashboard for monitoring attendance using React. Supported deployment in a production environment with 50–100 users. (Python, OpenCV, React, Javascript, Image Processing)
- **Face Mask Recognition system:** Developed a Face Mask Recognition system with a dataset of 2,000+ labeled images. Applied data augmentation techniques to enhance generalization under variable lighting and backgrounds. Trained custom CNN models and evaluated YOLOv3/YOLOv4, EfficientDet, SSD, RetinaNet, and Faster R-CNN, achieving 92%+ accuracy with real-time inference. Built a monitoring dashboard using React for live detection visualization... (Python, OpenCV, React, Javascript, Image Processing)

Web Developer - Outlier (AI Training)

[Remote]

Jan 2025 – Oct 2025

- **Helping cutting-edge generative AI models write better code:** Evaluate and optimize AI-generated code, write efficient solutions, create robust test cases, and provide clear, human-readable explanations of coding problems and solution approaches.
- **Simple perfume e-commerce website interface:** Built a perfume e-commerce site with responsive UI, product filtering, cart management, and order summary. Developed RESTful API for product listing, cart, and order processing using Node.js & Express. Integrated MongoDB for product/order storage, authentication, and admin product management. (JavaScript, ReactJS, HTML5, CSS, Node.js, Express, MongoDB, JWT)
- **Shareholder Management Dashboard:** Developed a web dashboard for managing shareholders and investment portfolios in a stock investment fund. Built with ReactJS + TailwindCSS, including interactive data tables, advanced filtering, sorting, and data visualization with Chart.js/Recharts. (ReactJS, TailwindCSS, Chart.js/Recharts)

SELECTED ML PROJECTS

3D Point Cloud Classification (Personal Project)

- Implemented preprocessing pipeline using Open3D/PCL: voxel downsampling, outlier removal, segmentation.

- Trained a PointNet model (PyTorch) for object classification on small-scale 3D datasets.
- Exported model to ONNX and performed inference using C++ ONNX Runtime API.
- Demonstrated understanding of robotics-oriented 3D ML workflows.

ROS2 Mobile Robot Obstacle Detection (Simulation-based)

- Implemented a basic computer vision pipeline using OpenCV (color thresholding / contour detection).
 - Subscribed to camera images from Gazebo, processed frames, and detected obstacles in front of the robot.
 - Published detection results as ROS 2 messages for downstream usage.
- (Tech stacks: ROS2, Python, OpenCV, Gazebo, Rviz2, TF2)

EDUCATION

Ho Chi Minh University of Technology

2017 – May 2022

Bachelor in Telecommunication and Electronics

Topic: 640

SKILLS & BACKGROUND KNOWLEDGE

Programming Languages: Python, SQL, JavaScript, Typescript, C++

Deep Learning Frameworks: Pytorch, TensorFlow, Keras

AI/ML: Deep Learning, Computer Vision, Image Processing, Data Augmentation

3D Processing: Open3D, PCL(familiar), Point Cloud Processing, PointNet (familiar)

Model Deployment: ONNX, ONNX Runtime, TensorRT (familiar), Model Optimization

Frontend: ReactJS, HTML5, CSS3, TailwindCSS

Backend: Node.js, Express, REST API

Database: MongoDB, PostgreSQL, MySQL

Dev Tools: Git, NPM, VSCode, Postman, Docker, Vercel

Automation: n8n

Cloud tools: AWS Cloud (AWS EC2, AWS S3, Lambda)

Robots: Experience with ROS2 (DDS QoS, Nav2, ...)

Others: WebSocket, JWT Authentication, API Integration